Quadratic Forms over the Full Nimber Field

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Abstract

Let K be a perfect field of characteristic 2 with surjective Artin–Schreier map. Every nonsingular finite-dimensional quadratic form over K is hyperbolic, so $W_q(K) = 0$. More generally, every quadratic form is classified by

$$\big(\dim V, \dim \operatorname{rad} B, \dim \operatorname{rad} Q\big),$$

where B is the polar form and $\operatorname{rad} Q = \{x \in \operatorname{rad} B : Q(x) = 0\}$. Conway's full nimber field On_2 satisfies these hypotheses. Consequently it has no additional transfinite Arf coordinate, and the Clifford–Brauer–Wall class of every nonsingular form is split. Over a declared finite nimsubfield, the ordinary Arf invariant remains meaningful, but it vanishes after scalar extension from $\mathbb{F}_{2^d}$ to $\mathbb{F}_{2^{2d}}$. The classification is formalized in Lean over set-sized fields; computations in the partial ordinal backend retain an explicitly finite-subfield-relative meaning.

1 Introduction

Nim addition and nim multiplication make the ordinals Conway's algebraically closed field On_2 of characteristic 2 [4, 8]. Quadratic-form invariants over finite nimber fields therefore raise a natural base-change question: what remains of the Arf invariant after extension to the full nimber field? The answer is that every nonsingular form becomes hyperbolic. A singular form retains only its polar rank and the distinction between zero and nonzero restriction to the polar radical.

The proof is a direct form of standard characteristic-2 quadratic-form theory [2, 3, 6]. Artin–Schreier surjectivity hyperbolizes every nonsingular plane, while perfection turns the quadratic restriction to the polar radical into the square of a linear functional. These two observations yield the complete normal form in Section 3. Section 4 explains the base change from finite nimber fields, and Section 5 records the Clifford consequence. Set-sized quadratically closed nimber fields are studied in [7, 9]; only the two displayed field hypotheses are needed for the classification itself.

2 Quadratic forms in characteristic two

Let K be a field of characteristic 2 and V a finite-dimensional K-vector space. A quadratic form is a map $Q : V \to K$ satisfying

$$Q(av) = a^2 Q(v), \qquad B(u, v) = Q(u + v) + Q(u) + Q(v),$$

where the polar form B is bilinear. The form B is alternating. We put

$$R = \operatorname{rad} B = \{v : B(v, w) = 0 \text{ for all } w \in V\}, \qquad \operatorname{rad} Q = \{r \in R : Q(r) = 0\}.$$

The quadratic form is *nonsingular* when $R = 0$. This is the standard characteristic-2 convention [6, Chapter II, Section 7].

Write

$$\wp : K \longrightarrow K, \qquad \wp(t) = t^2 + t.$$

For a nonsingular form and a symplectic basis $(e_i, f_i)_{i=1}^r$, its Arf class is represented by

$$\sum_{i=1}^{r} Q(e_i)Q(f_i) \pmod{\wp(K)}.$$

Over a finite field, absolute trace identifies $K/\wp(K)$ with $\mathbb{F}_2$. Over a field on which $\wp$ is surjective, the quotient is zero.

Let H denote the hyperbolic plane, with basis e, f satisfying

$$Q(e) = Q(f) = 0, \qquad B(e, f) = 1.$$

Lemma 2.1 (Splitting a symplectic plane)**.** *If $\wp$ is surjective, every nonsingular quadratic plane over K is hyperbolic.*

Proof. Choose e, f with $B(e, f) = 1$ and write $a = Q(e)$ and $b = Q(f)$. If $a = 0$, then e is isotropic. If $a \neq 0$, choose t with $t^2 + t = ab$ and put $v = (t/a)e + f$. Then

$$Q(v) = \frac{t^2 + t + ab}{a} = 0.$$

Thus in either case there is a nonzero isotropic vector v. Choose w_0 with $B(v, w_0) = 1$ and set $w = w_0 + Q(w_0)v$. Since $Q(v) = 0$,

$$Q(w) = Q(w_0) + Q(w_0)B(w_0, v) = 0, \qquad B(v, w) = 1.$$

Hence (v, w) is a hyperbolic basis. □

Proposition 2.2. *If $\wp$ is surjective, every nonsingular quadratic space of dimension $2r$ is isometric to $H^{\perp r}$. Consequently $W_q(K) = 0$.*

Proof. A nondegenerate alternating form has a symplectic plane. By Lemma 2.1, the restriction of Q to that plane is hyperbolic. Its polar-orthogonal complement is nonsingular, so induction splits off all r planes. Every nonsingular form is therefore Witt-trivial. □

The remaining information lies on the polar radical. Perfection is used precisely here.

Lemma 2.3 (Zero-polar forms)**.** *Suppose K is perfect and R is an s-dimensional space on which the polar form of Q vanishes. There is a unique linear functional $\ell : R \to K$ such that $Q(r) = \ell(r)^2$. Up to isometry,*

$$Q|_R \cong \begin{cases} 0^{\perp s}, & \ell = 0, \\ \langle x^2 \rangle \perp 0^{\perp (s-1)}, & \ell \neq 0. \end{cases}$$

Proof. For a basis $r_1, \dots, r_s$, perfection gives unique c_i with $c_i^2 = Q(r_i)$. Define $\ell(\sum_i x_i r_i) = \sum_i c_i x_i$. Since the polar form vanishes,

$$Q\left(\sum_i x_i r_i\right) = \sum_i x_i^2 Q(r_i) = \ell\left(\sum_i x_i r_i\right)^2.$$

Injectivity of Frobenius gives uniqueness. If $\ell \neq 0$, choose a basis in which ℓ is the first coordinate. □

3 The classification theorem

Theorem 3.1. *Let K be a perfect field of characteristic 2 with surjective Artin–Schreier map. Let Q be a quadratic form on V, let B be its polar form, and write*

$$\operatorname{rank} B = 2r, \qquad \dim \operatorname{rad} B = s.$$

Then exactly one of the following normal forms holds:

$$(V,Q) \cong \begin{cases} H^{\perp r} \perp 0^{\perp s}, & Q|_{\operatorname{rad} B} = 0, \\ H^{\perp r} \perp \langle x^2 \rangle \perp 0^{\perp (s-1)}, & Q|_{\operatorname{rad} B} \neq 0. \end{cases}$$

In the second case $s \geq 1$. Two such forms are isometric if and only if their triples

$$\big(\dim V, \dim \operatorname{rad} B, \dim \operatorname{rad} Q\big)$$

agree.

Proof. Choose a vector-space complement W to $R = \operatorname{rad} B$. The restriction $B|_W$ is nondegenerate: a vector in its radical is orthogonal to both W and R, hence lies in $W \cap R = 0$. Therefore

$$(V,Q) = (W, Q|_W) \perp (R, Q|_R).$$

Proposition 2.2 identifies the first summand with $H^{\perp r}$, and Lemma 2.3 gives the two radical summands. In the notation of that lemma, $\operatorname{rad} Q = \ker \ell$, so its dimension is s in the first case and $s-1$ in the second. These dimensions recover the normal form and are invariant under isometry. □

Corollary 3.2 (The full nimber field)**.** *Every nonsingular finite-dimensional quadratic form over* On_2 *is hyperbolic, and*

$$W_q(\mathrm{On}_2) = 0.$$

Every singular form over On_2 *has exactly one of the two normal forms in Theorem 3.1. In particular, no additional transfinite Arf invariant occurs.*

Proof. The field On_2 is algebraically closed of characteristic 2 [4, 8]. Algebraic closure makes both Frobenius and $x \mapsto x^2 + x$ surjective, so Theorem 3.1 applies. □

Remark 3.3 (Set-sized reduction)*.* Only finitely many coefficients and finitely many roots occur in the proof of a finite-dimensional instance. They lie in a set-sized algebraically closed subfield of On_2. Thus the corollary does not require performing linear algebra over a proper-class carrier.

4 Finite nim-subfields and base change

Let $K_d = \mathbb{F}_{2^d} \subset \mathrm{On}_2$. For a nonsingular quadratic form over K_d, the Arf class lies in

$$K_d/\wp(K_d) \cong \mathbb{F}_2,$$

with the isomorphism given by absolute trace. This is a relative invariant: it depends on the ground field over which changes of basis are allowed.

Proposition 4.1. *Scalar extension from K_d to K_{2d} induces the zero map*

$$W_q(K_d) \longrightarrow W_q(K_{2d}).$$

Consequently the directed colimit of the finite-field quadratic Witt groups inside On_2 *is zero.*

Proof. For $c \in K_d$, the polynomial $X^2 + X + c$ either splits over K_d or is an irreducible quadratic, in which case it splits over K_{2d}. Thus every class in $K_d/\wp(K_d)$ becomes zero in $K_{2d}/\wp(K_{2d})$. The finite-field classification by the Arf class [2, 5] gives the Witt-group statement. Every element at every finite stage therefore dies at a later stage, which is exactly the vanishing of the colimit. □

For example, the plane with $Q(e) = Q(f) = B(e, f) = 1$ is anisotropic over $\mathbb{F}_2$. If $t^2 + t + 1 = 0$ in $\mathbb{F}_4$, then $te + f$ is isotropic, so the same plane is hyperbolic over $\mathbb{F}_4$. Its Arf bit correctly records failure to split over $\mathbb{F}_2$; it is not invariant under unrestricted scalar extension.

5 Clifford consequence

Corollary 5.1. *For every nonsingular finite-dimensional quadratic form Q over On_2, the graded Morita class of $\mathrm{Cl}(Q)$ is split on the characteristic-2 Clifford–Brauer–Wall surface.*

Proof. By Corollary 3.2, $Q \cong H^{\perp r}$. Orthogonal sum corresponds to graded tensor product of Clifford algebras [6, Chapter IX]. For a hyperbolic basis e, f,

$$e^2 = f^2 = 0, \qquad ef + fe = 1,$$

so $\mathrm{Cl}(H)$ is a graded matrix algebra and is graded Morita-trivial. Hence $[\mathrm{Cl}(Q)] = 0$. □

This statement concerns Clifford algebras of nonsingular quadratic forms. For singular Q, the full Clifford algebra need not be graded central simple, so the singular radical flag in Theorem 3.1 is quadratic-form data rather than a Brauer–Wall coordinate.

6 Formalization and computational scope

The companion Lean development [1] proves the finite-dimensional classification over an arbitrary set-sized field with the hypotheses of Theorem 3.1. The division of work is:

- `Off.lean` proves the plane-splitting and zero-polar lemmas;
- `SymplecticBasis.lean` constructs a recursive orthogonal symplectic basis;
- `CharTwoClassification.lean` assembles the basis changes into explicit quadratic isometries and proves the converse classification by the two radical dimensions.

Thus the two normal forms, their direct-sum witnesses, and the converse classification by radical dimensions are kernel-checked. Algebraic closure is used only to specialize the field hypotheses. Lean does not encode the proper-class carrier of all nimbers or identify Mathlib field operations with Ogdoad's concrete arithmetic.

The executable `Ordinal` scalar is a checked partial representation of an initial Kummer region, not an algebraically closed field. For a finite metric, `ordinal_common_finite_subfield_degree` returns a common finite degree only when the necessary excess data and degree arithmetic are available. Arf and Brauer–Wall reports are then relative to that recorded finite field, as Proposition 4.1 requires. Rejection outside the verified arithmetic window does not define an additional quadratic-form class.

7 Conclusion

Perfectness and Artin–Schreier surjectivity remove every nonsingular characteristic-2 obstruction. Over On_2, all nonsingular forms are hyperbolic, $W_q(\mathrm{On}_2) = 0$, and their Clifford classes split. Singular forms retain exactly the polar- and quadratic-radical dimensions. Finite-field Arf bits are valid relative invariants, but every such class dies after a quadratic extension and hence in the directed limit. The full-field theorem and the finite executable interface therefore agree once their different ground-field scopes are kept explicit.

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